MAE4730/5730
Lecture 1

Introduction

Intermediate Dynamics

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Office Hours

HW 1 due Fri 9/3
Survey due Mon 8/30
Canvas open today.
Ed Discussion.

Concerned with the motion of particles and rigid objects based on the relationship between extensive properties and applied forces.

Goal Given a system; you should be able to

* Write Equations of Motion (EoM) { Newton-Euler / Euler-Lagrange
* Generate a numerical Solution
* Animate it
* Make engineering deduction

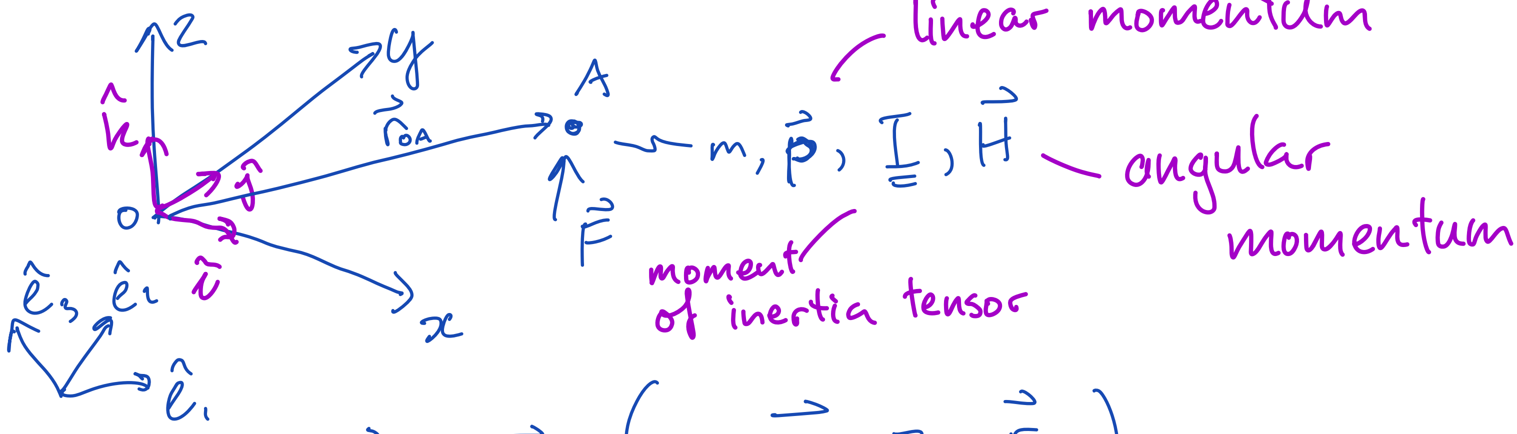
Any 2D system, or 3D single rigid object.

Homework Due Fridays 6pm (20%, of course, lowest 2 dropped)
↳ Gradescope & Canvas

2 Prelims, 1 Final Exam, TBA (60% total)

Final Project (20%, Individual Submission)

General Notation & FBDs



$$\vec{r}_{OA} \equiv \vec{r}_{A/O} \left(\equiv -\vec{r}_{AO} \equiv -\vec{r}_{O/A} \right)$$

I Kinematics

Geometry in motion

$$\begin{cases} \vec{v} = \frac{d}{dt}\vec{r} = \dot{\vec{r}} \\ \vec{a} = \frac{d}{dt}\vec{v} = \dot{\vec{v}} = \ddot{\vec{r}} \end{cases}$$

II Laws of Motion

First Law: Conservation of Momentum

Second Law: $\vec{F} = m\vec{a}$

$$\boxed{\sum_{\text{external}} \vec{F}_i = \dot{\vec{p}}}$$

Linear Momentum Balance (LMB)

Also: Conservation of Angular Momentum

$$\boxed{\sum_{\text{external}} \vec{M}_i = \dot{\vec{H}}}$$

Angular Momentum Balance (AMB)

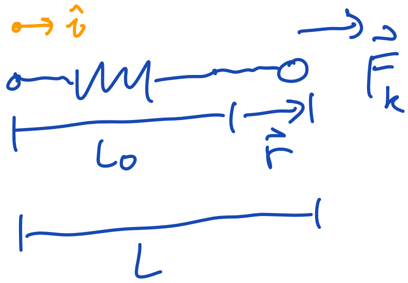
III Constitutive Laws

Finding $\vec{F}$ in terms $(\vec{r}, \vec{v}, t)$

Kinematics $\Longleftrightarrow$ Kinetics

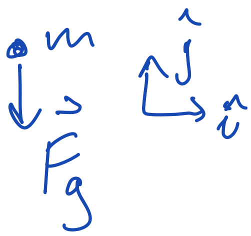
Examples

- Spring



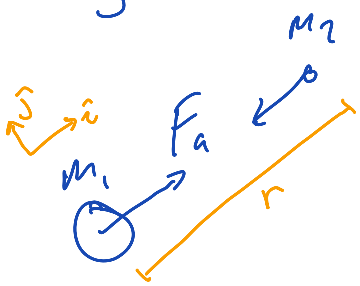
$$\vec{F} = (L - L_o)k\left(\frac{-\vec{r}}{|r|}\right)$$

- Near Earth Gravity



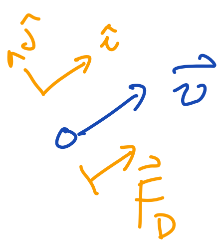
$$\vec{F}_g = -mg\hat{j}$$

- Inverse Square Law Gravity

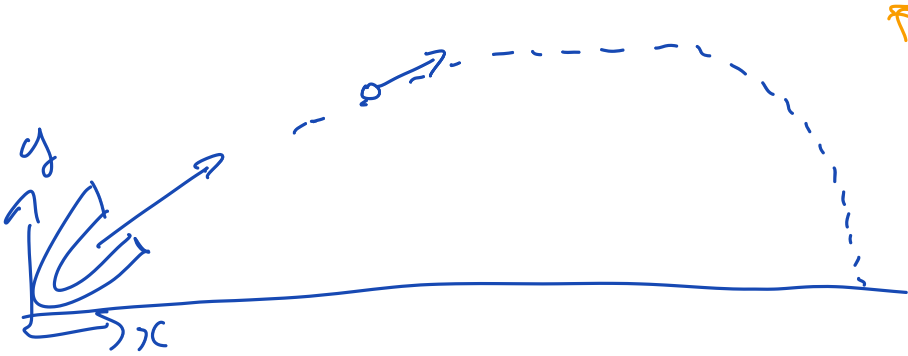


$$\vec{F}_G = \frac{G m_1 m_2}{r^2}\left(\frac{\vec{r}}{|r|}\right)$$

- Air Drag



$$\vec{F}_D = c v^2 \left(\frac{-\vec{v}}{|v|}\right)$$



Post Lecture Note: Always define a coord. system to avoid confusion.

Matlab grader $\rightarrow$ symbolic toolbox?